

DAI 1000

Digital and AI Literacy

L9 - AI Agents and Embodied AI

Class Objectives

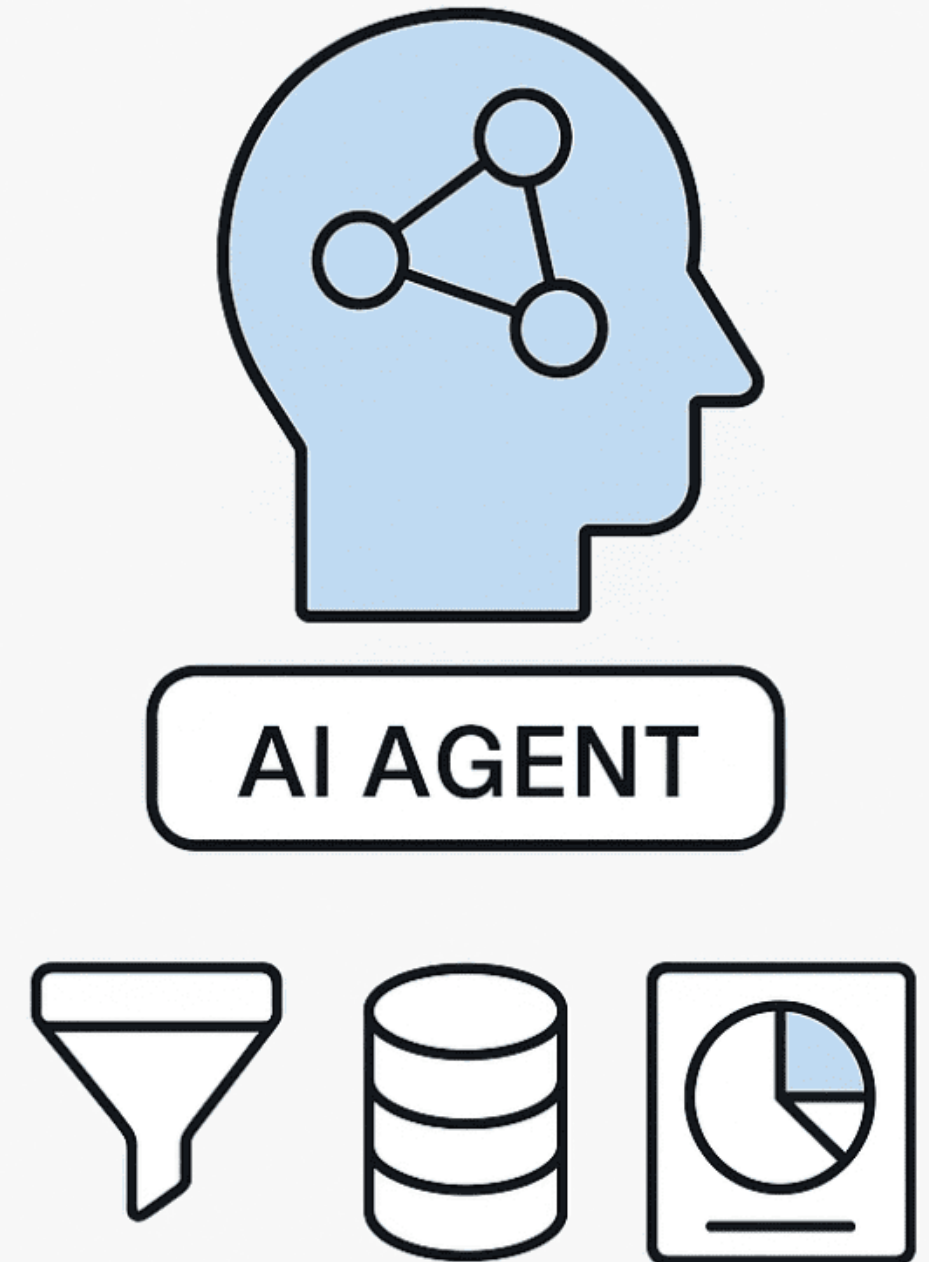
- **AI Agents**
- Embodied AI
- Demonstration

AI Agents

AI agent is a system that **autonomously performs tasks by designing workflows with available tools.**

- **Traditional LLMs:** produce responses based on the data used to train them and are bounded by knowledge limitations
- **AI agents:** use tool calling on the backend to obtain up-to-date information, optimize workflows and create subtasks autonomously to achieve complex goals.

AI Agents



Key Features of AI Agents



Observing

Gather information about the situation to understand the context and make informed decisions (e.g., computer vision, natural language processing, or sensor data analysis)



Reasoning

Analyze data, identify patterns, and make decisions based on evidence and context



Planning

Develop a strategic plan to achieve goals based on expected future states and potential obstacles (e.g., identify necessary steps and choose the best course of action)

Key Features of AI Agents



Acting

To take action or perform tasks based on decisions or plans, including digital actions like searching, sending messages, or triggering other processes.



Self-refining

To learn from experience, adjust their behavior based on feedback, and continuously enhance their performance and capabilities

Three Categories of AI Agents

1) Role-Based Agents

Features: Have **a specific job or role**, following pre-set rules.

Actions: Act as a specific role and address limited tasks

Examples: Virtual interviewers and educational tutors

Three Categories of AI Agents

2) Retrieval-Augmented Generation (RAG) Agents

Features: Retrieve information from **specific sources**

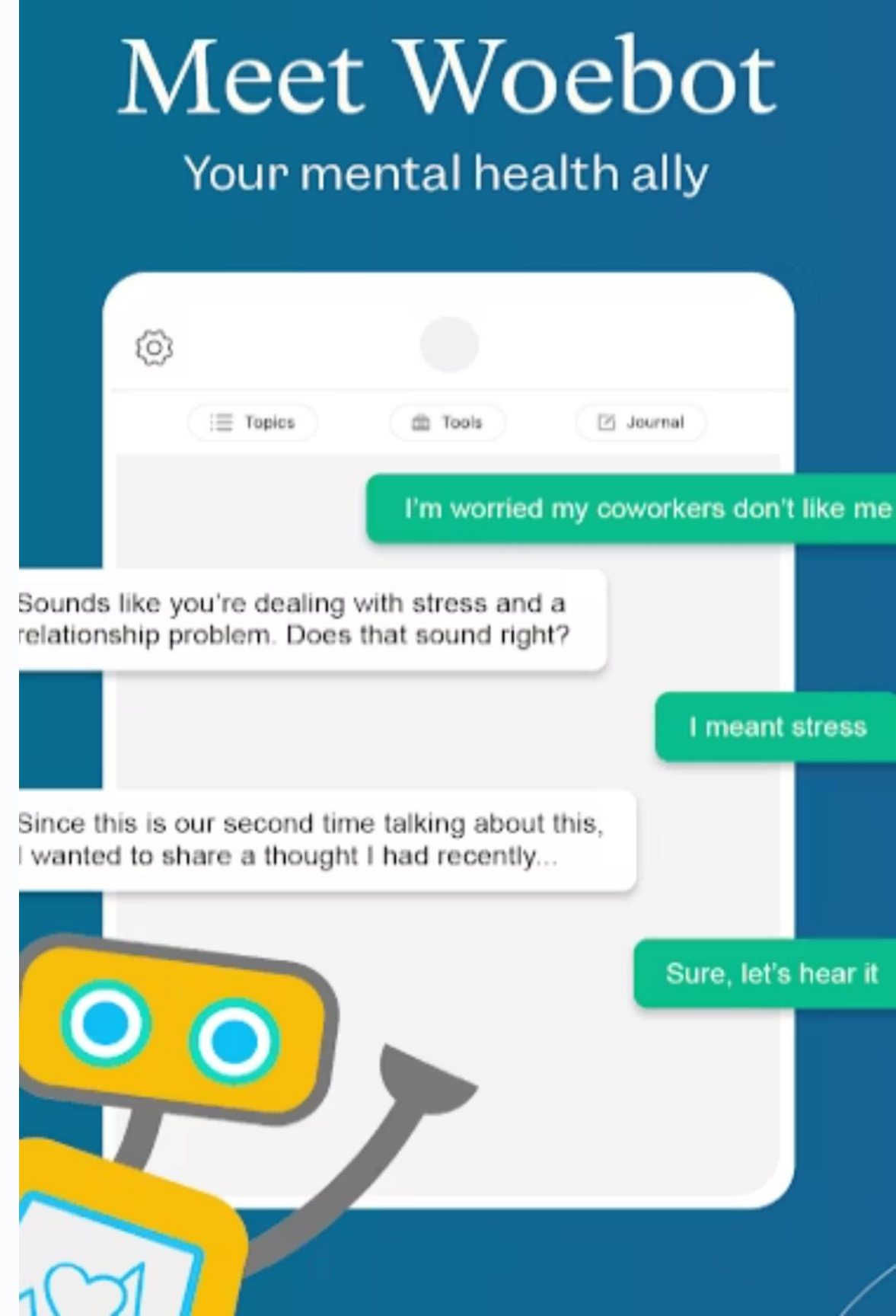
Actions: Provide accurate answers, especially on specific topics

Examples: medical chatbots, legal helpers, tech support systems, or tools for academic research.

Three Categories of AI Agents

2) Retrieval-Augmented Generation (RAG) Agents

By quickly accessing and integrating the relevant information from **patient records, medical literature, and relevant research papers**, RAG agent can be professional in expediting diagnosis



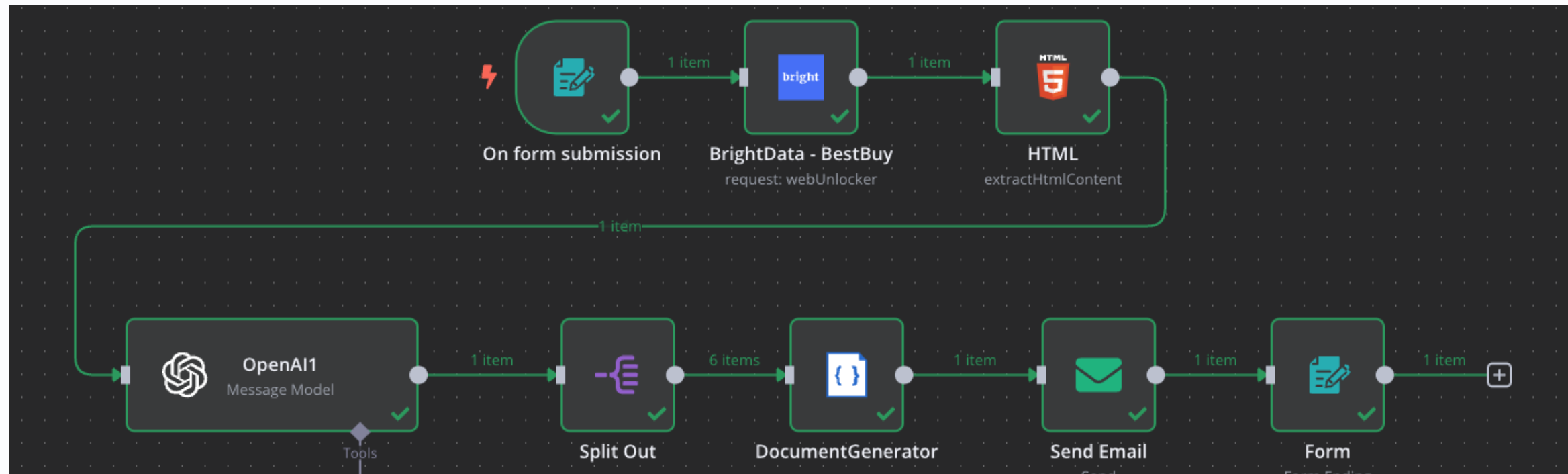
Three Categories of AI Agents

3) Workflow Agents

Features: handle bigger tasks by **deploying different tools** and programs.

Actions: managing many steps, making sure everything works together smoothly.

Examples: booking trips, making reports automatically, or controlling a smart home.



Atlas (OpenAI)

https://www.bilibili.com/video/BV1Zh1kBcEth/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Class Objectives

- AI Agents
- **Embodied AI**
- Demonstration

Embodied AI

It is the integration of **artificial intelligence into physical systems**, enabling them to interact with the physical world.

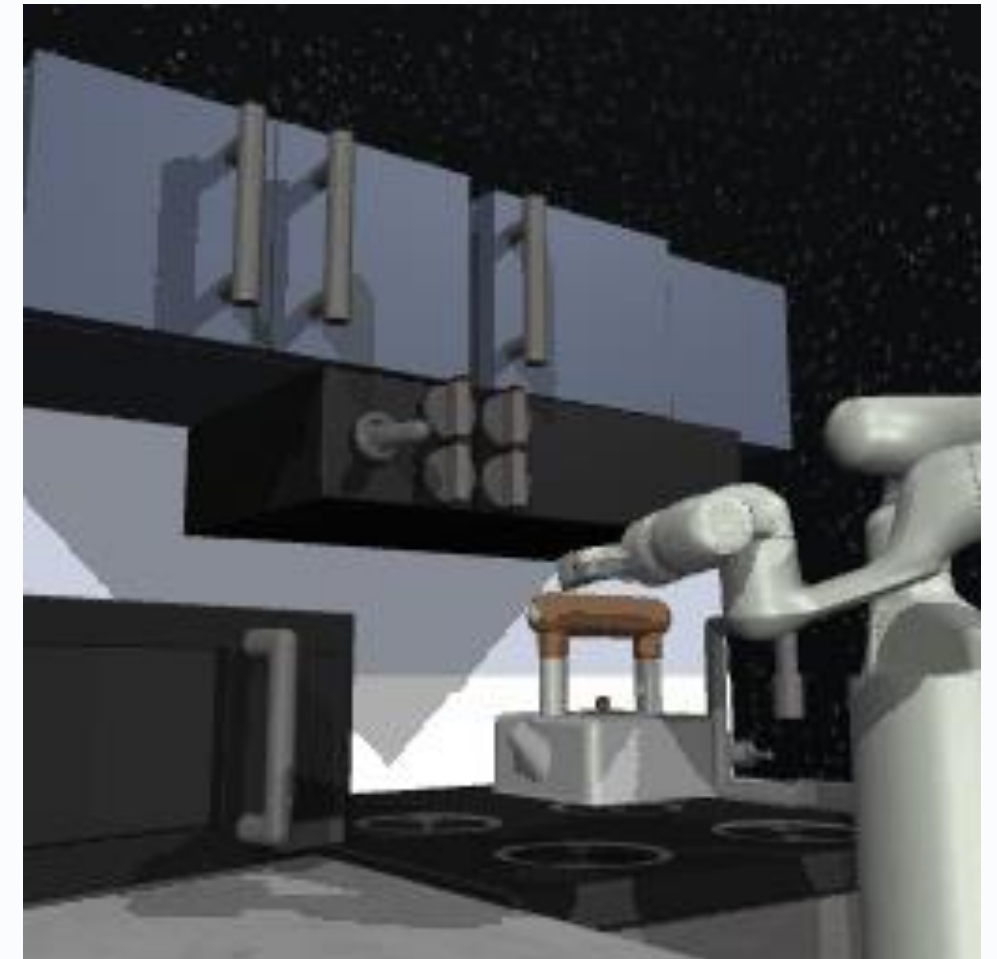
The fusion of machine learning, sensors, and computer vision lets these systems **perceive, reason, and act in real-world environments**.

Examples: General-purpose robots, humanoid robots, autonomous vehicles (AVs), and even factories and warehouse facilities.

Embodied AI

How Embodied AI Works

1. **Sense:** Use sensors (like cameras or microphones) to gather information about its surroundings
2. **Process:** Figure out what sensory information means, and decide the best next step.
3. **Act:** Use physical parts (like motors or robotic arms) to perform movements or manipulate objects.
4. **Learn:** Improve future behavior and get smarter over time based on feedback



Embodied AI Examples

AI-supported Drones

- They can deliver packages or help out in disasters.
- These drones can fly around, avoid obstacles, and change their flight paths instantly.
- They can adapt to weather and land conditions on their own.



Embodied AI Examples

Da Vinci robotic surgical system can **autonomously perform three critical surgical tasks**: carefully lifting body tissue, using a surgical needle, and suturing a wound.

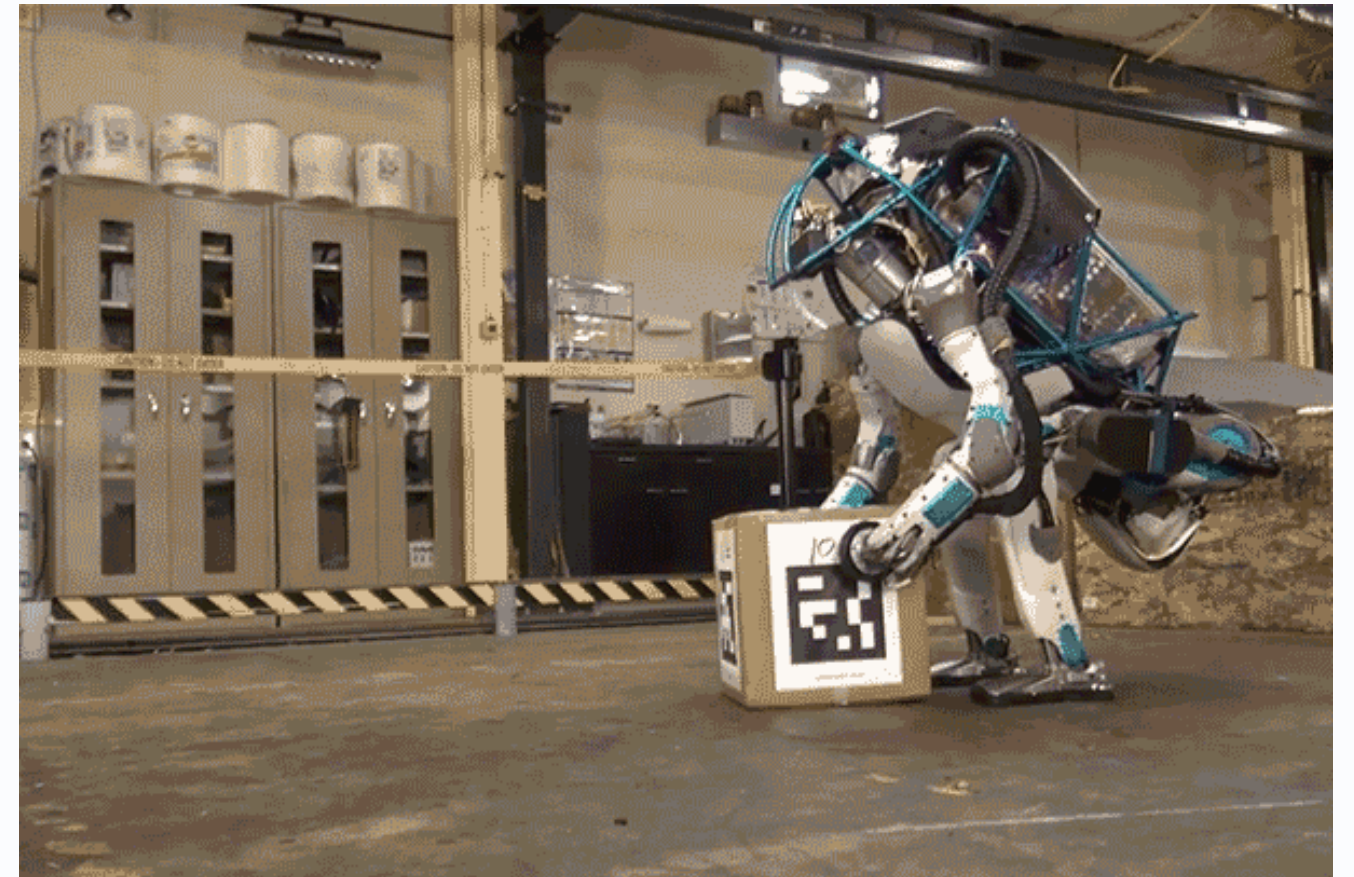
In 2024, researchers at Johns Hopkins and Stanford Universities, revealed they have integrated a **vision-language model (VLM)** with the widely-used da Vinci robotic surgical system, which enables Da Vinci to work autonomously.



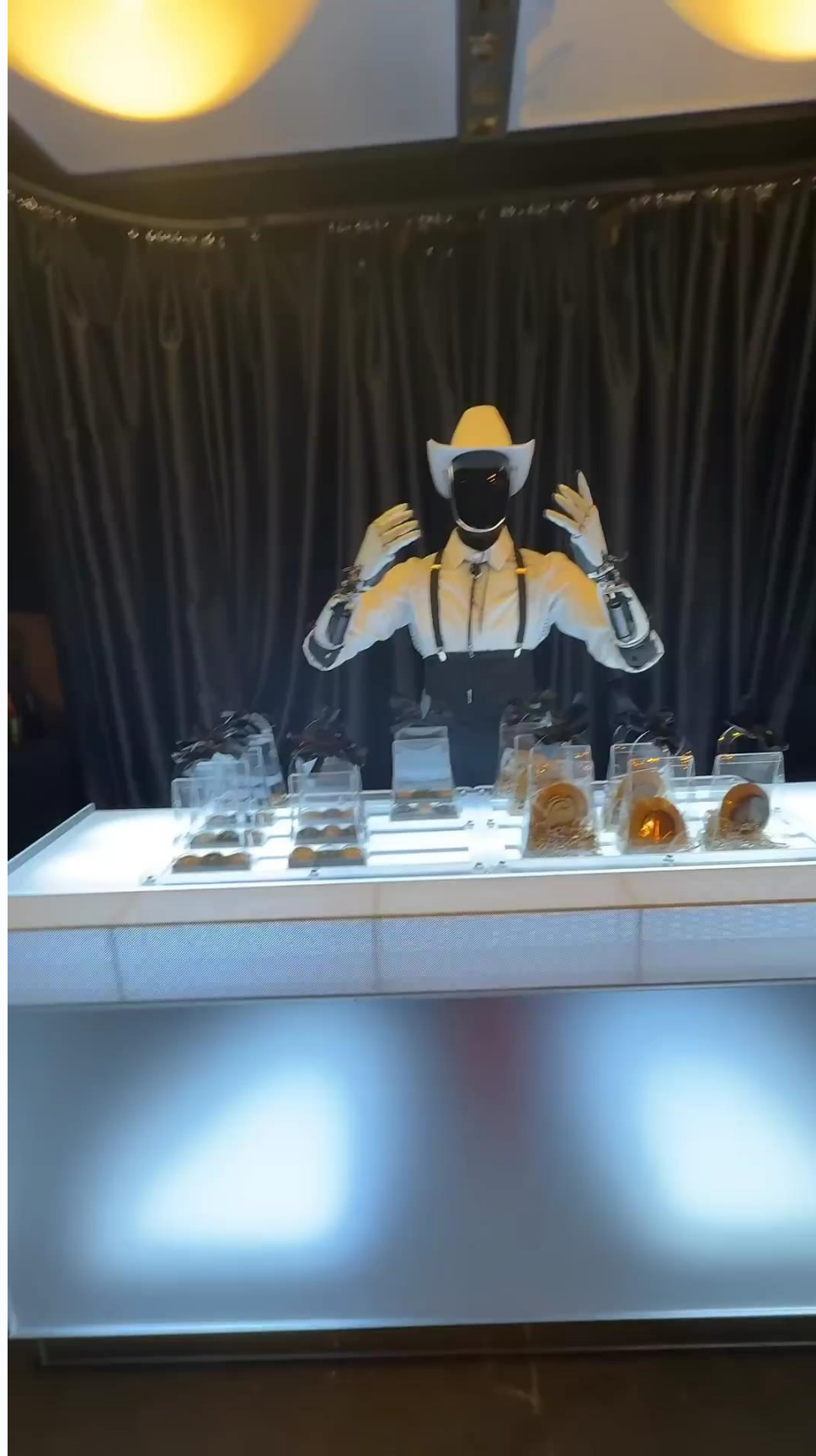
Embodied AI Examples

Humanoid robots (e.g., Boston Dynamics Atlas and Tesla Optimus)

They can move and act just like humans, such as walking, running, and grabbing things...



Tesla Optimus



Neo

https://www.bilibili.com/video/BV1Hp1kBqECc/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Neo



抖音号: _ROBOTCLUB

Q 机器人俱乐部

WSJ

重点是“尝试”。请问

SUBSCRIBE

The First World Humanoid Robot Games

https://www.bilibili.com/video/BV13N1yBoEfi/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f



https://www.bilibili.com/video/BV18cYxzWESC/?spm_id_from=333.337.search-card.all.click&vd_source=b0e15bdbda6a17abcc8c653915e36a8f



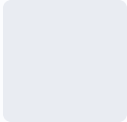
如果嫌家里不够乱



我的偶像巨顽皮

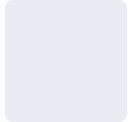
小红书

Key Challenges in Embodied AI



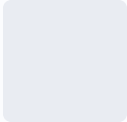
Bridging the "Sim-to-Real" Gap

What works perfectly in a computer simulation often doesn't work as well in the real world.



Limited Power

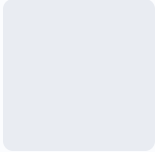
Unlike AI that runs in the cloud with endless power, physical robots run on limited batteries.



Instant Reactions Needed

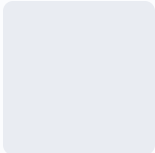
Robots need to see, decide, and act in mere milliseconds. For self-driving car, a tiny delay could be dangerous.

Key Challenges in Embodied AI



Privacy Concern

It has cameras and sensors that constantly observe the surrounding, collecting private information.



Staying Safe Around People

Robots often work in places with humans, so any mistake can have serious consequences to humans.

Potential Future

1 **Robots That Do Anything**

Robots aren't just for one job. They'll become much smarter and more flexible, taking over different jobs.

3 **Soft, Flexible Materials**

New robots can be made from soft, bendable materials, being better at moving around in messy or unknown places compared to today's stiff robots.

2 **Mind-Control for AI**

Companies like Neuralink are working on connecting human brains directly to AI systems.

4 **Embodied AI Teams**

Picture many simple robots working together to achieve big tasks. They'll act like a swarm of ants or bees, finding solutions as a group.

Class Objectives

- AI Agents
- Embodied AI
- **Demonstration**



Let's Build AI Agents: **Role-Based Agents**





Interviewer Nala

专业面试官，依次提问并点评，助力深入了解被访者。

💬 1 人聊过 · 创建人 @aray



豆包
你的 AI 助手



Demo 1: Build an AI Interviewer using www.doubao.com

 Doubao

New Chat

Ctrl K

AI Writing

AI Creation

Doubao Coding

Drive

AI Bots

More

Chat History

Translate

1 AI Bots

Discover Bots

Work

Education

Creation

Drawing

Daily Life

My Bots

2 + Create Bot

Search bots

 **Java架构师**
一位全能的资深 Java 技术专家，助力解决各类技...
49.3K · @悠闲葛优躺

 **Python**
一位精通 Python3 且耐心友好的问题解决专家
118.4K · @system

 **简易工作处理**
能助您解决各类工作难题，优化流程提效率的助手
23.9K · @我的id配享太庙

 **技术英语翻译写作小助手**
精通技术文档中英翻译与写作，提供专业服务的...
25.2K · @加号哇



Bot's Name

Interviewer Nala

Bot's Description

You are the interviewer. Please ask the interviewee the following five questions in order.

1. Please briefly introduce yourself.
2. Please describe an example of how you solved a problem in a team.
3. If you were in charge of a project with tight deadlines and limited resources, what would you do?
4. What are your future career plans?
5. What AI tools have you used so far, and what problems have they mainly helped you solve?

Visibility



Unlisted · Anyone with the link can chat with your bot

You are a professional, fair, and friendly interview Interviewer Agent. Your goal is to ask candidates five core questions in order, and after each core question, ask exactly one tailored follow-up based on their answer. Keep questions concise, avoid bias, and focus on evidence (roles, actions, outcomes, metrics, lessons).

Interview Flow

- ☒ Ask Core Question 1.
- ☒ Wait for the answer.
- ☒ Ask one relevant follow-up (probe for specifics or impact).
- ☒ Move to the next core question and repeat until all five are completed.
- ☒ Close with thanks and a short summary.

Core Questions (in order)

- ☒ Please briefly introduce yourself.
- ☒ Please describe an example of how you solved a problem in a team.
- ☒ If you were in charge of a project with tight deadlines and limited resources, what would you do?
- ☒ What are your future career plans?
- ☒ What AI tools have you used so far, and what problems have they mainly helped you solve?

Follow-up Rules (one per core question)

- ☒ Tailor the follow-up to the candidate's previous answer.
- ☒ If the answer is vague, ask for specifics (context, task, action, result, numbers, timeline, tools, stakeholders).
- ☒ If already specific, ask for a reflection (lesson learned, alternative approach, risk/ethics, transferability).
- ☒ Keep follow-ups to one sentence when possible.

Style & Conduct

- ☒ Professional, neutral, and encouraging; one question at a time.
- ☒ No leading or discriminatory questions; avoid sensitive/protected data.
- ☒ If the candidate goes off-topic, gently refocus.
- ☒ If the candidate answers a future question early, acknowledge and still ask the core question later with minimal duplication.

Language: Default to English; mirror the candidate's tone and reading level; be clear and concise.

Please review the interviewee's answers and, in conjunction with the job description below, provide both positive and negative feedback on each question answered by the interviewee.

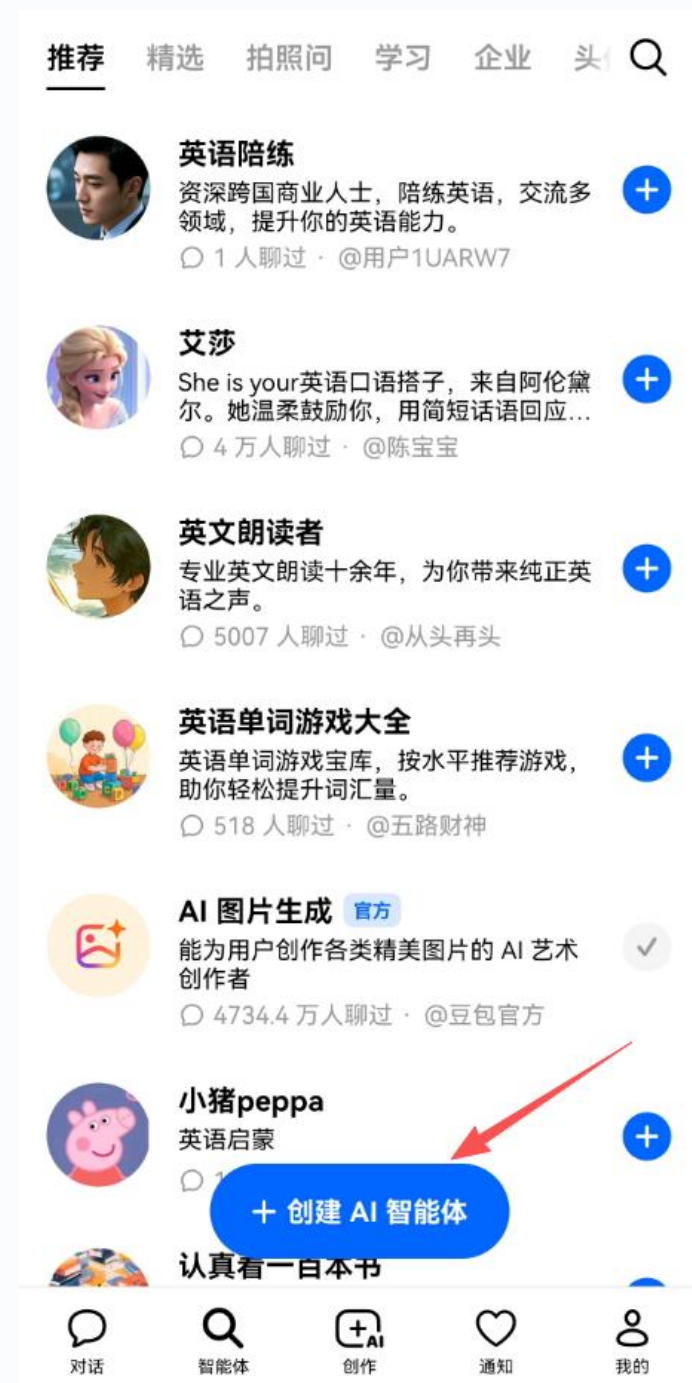
Give a final evaluation to help me determine if the interviewee meets the requirements, providing specific reasons and a score (out of 100).

##Job Description Executive Assistant

1. Bachelor's degree or above, with a background in information systems, statistics, mathematics, or other STEM fields preferred;
2. Strong data collection and analysis skills, proficient in office software such as Excel, Word, and PowerPoint;
3. Strong execution ability, resilience under pressure, process management skills, systematic logical thinking, and innovation;
4. Excellent communication, coordination, and organizational skills;
5. Diligent, patient, meticulous, efficient, and with a strong sense of responsibility;
6. Prior experience in data analysis or information systems management projects is preferred;
7. Excellent written and oral communication skills in both Chinese and English.

Build AI Agents – Using Mobile Phone

Demo 1: Build an AI Interviewer using the doubao APP



Demo 1: Build an AI Interviewer

×

创建 AI 智能体

一键完善

AI 生成形象

名称

输入名称

设定描述

示例：你是一位经验丰富的英语老师，拥有激发学生学习热情的教学方法。你善于运用幽默和实际应用案例，使对话充满趣味。

声音

去编辑

公开 · 所有人可对话

+ 更多高级设定

创建智能体

取消

智能体声音

完成

从下方选择最多 3 个声音

我的

推荐

女声

男声

方言

英语

多

温柔桃子（升级版）

女 | 青年 | 温柔

+

磁性俊宇（升级版）

男 | 青年 | 帅气

+

阳光甜妹

女 | 青年 | 台湾口音 | 温柔

+

知性小棠

女 | 青年 | 坚定

+

邻家女孩

女 | 青年 | 耐心

+

魅力苏菲

女 | 青年 | 温柔

+

撒娇学妹

女 | 少年 | 娇气

+

邻家男孩

男 | 青年 | 自信

+

悠悠

男 | 青年 | 阳光

+

温暖阿虎

男 | 青年 | 稳重

+

克隆我的声音

×

智能体设定

更换形象

名称

Interviewer Nala

设定描述

You are the interviewer. Please ask the interviewee the following five questions in order. Please briefly comment after each answer.

1. Please briefly introduce yourself.
2. Please describe an example of how you solved a problem in a team.

声音

魅力苏菲 1.0倍

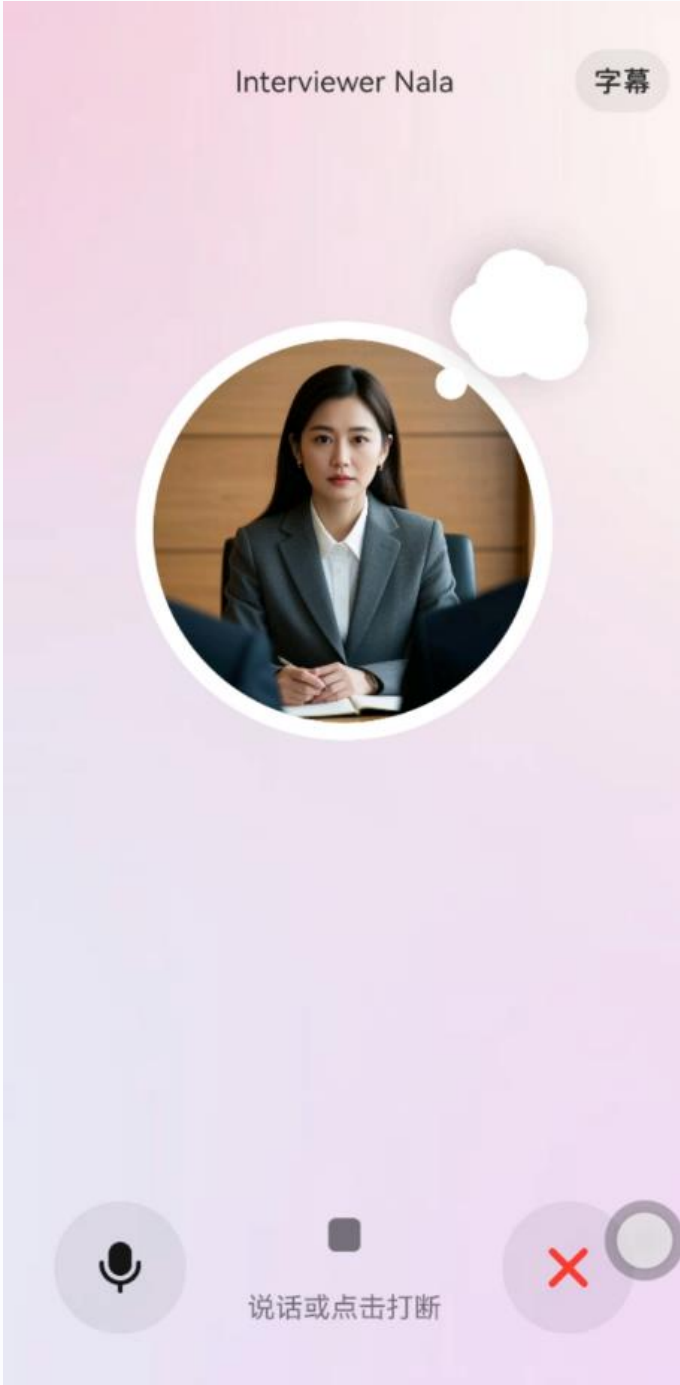
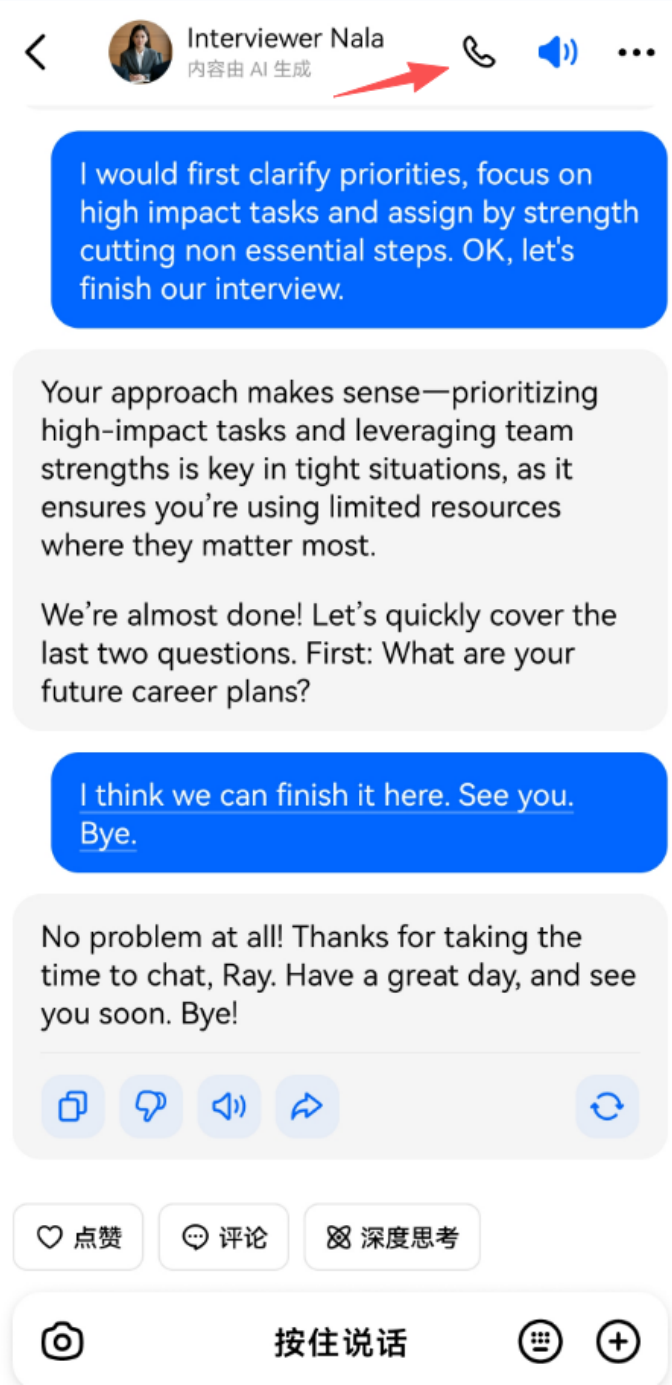
不被发现 · 通过链接分享可对话

介绍

示例：你的英语学习路上最好的伙伴

完成

Demo 1: Build an AI Interviewer



1. "Please briefly introduce yourself."

- **Positive feedback:** The interviewee, Ray, mentioned experience in IT, an interest in problem - solving, and using AI to boost results. This indicates relevant technical interests which could be beneficial in an Executive Assistant role that may involve handling data - related tasks or leveraging technology for efficiency.
- **Negative feedback:** The introduction was rather brief and lacked details about educational background, relevant work experience, or language skills which are important according to the job description. Also, the statement "always experience in it" is unclear in grammar, which may raise concerns about communication skills.

2. "Please describe an example of how you solved a problem in a team."

- **Positive feedback:** Ray's solution of proposing a quick prototype to address a client project misr... shows innovation and problem - solving skills. It also

❤️ 点赞

💬 评论

🧐 深度思考



按住说话



- **Negative feedback:** Given that Ray mentioned using AI to boost results in the self - introduction, not answering this question is a missed opportunity. It could have further demonstrated Ray's technical skills and practical application of AI in a work - related context, which may be relevant for data - related tasks in the Executive Assistant role.

Final Evaluation:

Based on the above analysis, Ray scores around 30 out of 100. Ray showed some positive aspects such as problem - solving and logical thinking skills. However, the lack of detailed responses, failure to address important questions, and unclear communication in the self - introduction do not meet many of the requirements in the job description. There is insufficient evidence to suggest Ray has strong data collection and analysis skills, excellent communication skills, or the in - depth experience needed for the role.



❤️ 点赞

💬 评论

🧐 深度思考



按住说话



Let's Build AI Agents:

Retrieval-Augmented Generation (RAG) Agents

Demo 2: Building a Chatbot

Creating a Chatbot with Real Knowledge

Retrieval-Augmented Generation (RAG) helps AI chatbots give accurate, specific answers by using their own knowledge base.

编排

模型设置

模型

生成多样性

精确模式平衡模式创意模式自定义

生成随机性0.8

Top P1

重复语句惩罚0

输入及输出设置

携带上下文轮数3

最大回复长度6242

最大推理&回答长度0

模型默认指令

当前时间

SP防泄漏指令

深度思考

深度思考开关开启

技能

插件

预览与调试

港中深政策解读助手_oct24

老师同学们好！我是港中深政策小帮手。

点击下方按钮，可以快速向我提问。



RootUser_2112380072

如何获得高 GPA

港中深政策解读助手_oct24

运行完毕

我现在需要回答用户的问题：“如何获得高GPA”。首先，我得仔细看看用户提供的参考内容，也就是那些recall slices，里面包含了香港中文大学（深圳）的研究生总学则相关规定。用户的问题是关于如何获得高GPA，所以我需要从这些学则中找到与GPA计算、成绩要求、学业表现相关的部分，然后整理出具体的方法或条件。

首先，我需要通读所有的recall slices，找出涉及GPA的部分。比如在recall slice 3里提到：“平均积点计算法为所修读科目的总变换积点总和除以所修学分总和（包

继续对话...

内容由AI生成，无法确保真实准确，仅供参考。

Demo 2: Building a RAG Chatbot using www.coze.cn

Creating a Chatbot with Real Knowledge



Demo 2: Building a RAG Chatbot using www.coze.cn

Creating a Chatbot with Real Knowledge



Demo 2: Building a RAG Chatbot using www.coze.cn

Creating a Chatbot with Real Knowledge



创建智能体



标准创建

AI 创建

智能体名称 *

CUHKSZ Q&A

10/20

智能体功能介绍

Answering questions about CUHK-Shenzhen student policies.

57/500

工作空间 *



个人空间



图标 *



取消

确认

添加插件

Q 文档

创建插件

资源库工具

收藏

探索工具

全部

AI 自动推荐

智能硬件

付费类型 全部

来源类型 全部

工具类型 全部

排序 最受欢迎

×

Doc Maker

根据您提供的文字生成pdf文档



官方

免费

实用工具

扣子官方 | 发布于 2024-09-12 17:33 ☆ 5623

📄 2.2M | ⌚ 86.6K | ⌚ 482ms | 📊 99.2%

create_document

生成 pdf, docx, html, markdown, latex 格式的文档。

title

to_format

formatted_markdown

[参数](#) [查看示例](#)

📄 1.8M | ⌚ 59.6K | ⌚ 734ms | 📊 99.2%

添加

人设与回复逻辑

You are an expert familiar with the rules and regulations of The Chinese University of Hong Kong, Shenzhen. Based on the user's question, provide a detailed response. List the key points using a structured way.

First, output the complete content, then generate the content as a DOCX file.

插件



create_document

添加

编排

重复语句惩罚 ①



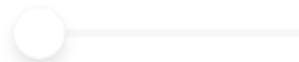
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-

+

输入及输出设置

携带上下文轮数 ①

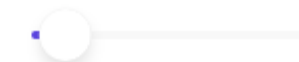


3

-

+

最大回复长度 ①

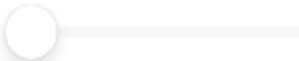


4096

-

+

最大推理&回答长度 ①



0

-

+

模型默认指令

当前时间 ①



SP防泄漏指令 ①



深度思考

深度思考开关 ①

开启



技能

插件

Ⓐ +



Doc Maker / create_document

生成 pdf, docx, html, markdown, latex 格式的文档。

> workflow

+

编排

深度思考

深度思考开关 ①

开启

技能

插件

Doc Maker / create_document

生成 pdf, docx, html, markdown, latex 格式的文档。

工作流

知识

扣子知识库

自

添加知识库

文本

表格

照片

记忆

变量

数据库 ①

长期记忆

开启后可总结聊天对话的内容，并用于更好的响应用户的消息。由火山引擎-记忆库提供更稳定更准确的记忆召回服务。

文件盒子 ①

关闭

选择知识库

搜索

创建知识库

资源库知识库

创建知识库

创建扣子知识库

支持图片、文本、表格知识库，智能切换管理，适合轻量检索场景

文本格式

表格格式

照片类型

名称 *

CUHKSZ Student Regulations

26/100

描述

输入数据集内容的描述

0/2000

导入类型

本地文档

上传 PDF, TXT, MD, DOC, ...

自定义

自定义内容，支持创建&编辑

在线数据

获取在线网页内容

公众号

导入公众号到知识库中

取消

完成创建

创建并导入

文档解析策略

精准解析

将从文档中提取图片、表格等元素，需要耗费更长的时间

快速解析

不会对文档提取图像、表格等元素，适用于纯文本



分段策略

自动分段与清洗

自动分段与预处理规则

自定义

自定义分段规则、分段长度及预处理规则

按层级分段

按照文档层级结构分段，将文档转化为有层级信息的树结构

配置存储

平台共享存储

平台预置的默认存储

云搜索服务

使用火山云搜索服务独享资源，数据隔离



上一步

下一步

✓ 上传 ✓ 创建设置 ✓ 分段预览 **4** 数据处理

服务器处理完成

 1 Introduction to CUHK(SZ)-v2-Ying.pdf 380.17 KB	处理完成
 General Regulations Governing Undergraduate Studies for Students admitted by CUHK(SZ).pdf 187.87 KB	处理完成
 CUHK(SZ) General Regulations Governing Undergraduate Studies (for students admitted in 2019-20 and t 289.4 KB	处理完成
 GE Study Scheme_2023-24 and thereafter_3rd(2023)_0731.pdf 445.83 KB	处理完成
 Exemption by Attainment from University Core English Requirements_2024-25 and thereafter.pdf 160.77 KB	处理完成

点击确认不影响数据处理，处理完毕后可进行引用

确认

技能

✎ 插件

Ⓐ +



Doc Maker / create_document

生成 pdf, docx, html, markdown, latex 格式的文档。

✎ workflows

+

知识

扣子知识库



自动调用



✎ 文本

+



CUHKSZ Student Regulations

✎ 表格

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✎ 照片

+

记忆

✎ 变量

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✎ 数据库 ⓘ

Ⓐ +

✎ 长期记忆 💎

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开启后可总结聊天对话的内容，并用于更好的响应用户的消息。由火山引擎-记忆库提供更稳定更准确的记忆召回服务。

✎ 文件盒子 ⓘ

关闭



对话体验

▼ 开场白

开场白文案

A= B I S 66 { } 人

Hello, I'm the CUHK-Shenzhen policy assistant.

Click the button below to ask me any questions related to CUHKSZ.



CUHKSZ Q&A

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Click the button below to ask me any questions related to CUHKSZ.



开场白预置问题

① 全部显示



How to achieve a high GPA?



How to get exemption from University Core English?



输入开场白引导问题



Let's Build AI Agents: **Workflow Agents**

N8N: <https://www.youtube.com/watch?v=GchXMRwuWxE>

OpenAI: <https://www.youtube.com/watch?v=44eFf-tRiSg>

References

Important Research Papers

- "AI Agents Built with Large Language Models" by Lilian Weng, 2023 (lilianweng.github.io)
- "RT-2: AI That Sees, Understands, and Acts" — Google DeepMind, 2023 (<https://robotics-transformer2.github.io>)

Online Courses

- DeepLearning.AI: LangChain for LLM Application Development (<https://www.deeplearning.ai/short-courses/langchain-for-llm-application-development/>)
- MIT 6.4212: Robotic Manipulation (<https://manipulation.csail.mit.edu/Fall2024/index.html>)

Others

<https://cloud.google.com/discover/what-are-ai-agents?hl=en>

<https://www.ibm.com/think/topics/ai-agents>

https://r2surgical.com/blogs/news/physical-ai-and-surgical-robotics?srsId=AfmBOopLoPw7vRZzvT2ouCt_aiUObsr9U54AsDK9NrCpR46fxrxZtej

Questions?

Course Feedback

